

Lab 5: Survey Data Analysis

Overview

This lab introduces the essential principles of **survey data analysis**. Public health data often comes from complex surveys like NHANES, which don't sample people randomly — they use a strategic design with **weights, strata, and clusters** so the sample represents the entire U.S. population. Ignoring these features can badly bias your results.

Using data from Flegal et al. (2016), you will work through the difference between a naive analysis and a methodologically sound survey analysis: creating a survey design object, generating weighted descriptive statistics, and fitting survey-weighted regression models to replicate published findings. Your TA will walk through the worked solution during the session.

This is the workflow you will reuse in **P4** on your own locked paper.

Materials

- **Exercise instructions:** Survey Data Analysis
- **Worked solution:** The R codes

Before you come to lab, install the package this exercise needs:

```
install.packages("survey")
```

Your task

1. Follow along as the TA explains the worked solution.
2. Run the R Markdown file yourself, as a group, on your own laptop.
3. Put your **group name and the first names of everyone present** in the `author:` field at the top of the file.
4. Add `sessionInfo()` in a final code chunk.
5. Knit to **PDF or HTML** and submit that one file before the lab ends.

No written explanations are required. If you get stuck, ask the TA in the room — that is what the session is for.

Deliverable

- **One file per group:** the knitted PDF or HTML.
- **Due:** end of the lab session (Thu Oct 15).

Grading: Complete / Incomplete

You receive **Complete** if the document knitted without errors, your group name and members are in it, and it was submitted before the lab ended. Only members present in the room receive credit.

Where this lab fits

This lab is one step in a weekly chain: Tuesday's lecture sets up the method, Thursday's lab runs it on fixed teaching data, and your increment applies it to the paper your group locked at M0.

	Document	How this lab connects
This week's lecture	Week 6 — Complex survey design (weights, strata, PSU)	Tuesday showed why a complex survey is not a simple random sample, and what weights, strata and PSUs are actually doing. This lab makes the contrast concrete: a naive unweighted fit beside a design-aware one, with the attenuation and the CI widening tabulated side by side. The same material as slides, with speaker notes. Press S for notes, F for fullscreen.
Slide deck	Week 6 deck	
Your increment	P4 — Design-aware estimate	Apply the same workflow to <i>your</i> paper's headline estimate: declare and justify the weight you chose, report the naive-versus-weighted contrast, and write the one-line design statement your paper omitted.
Worked example	P4 example — its code	A completed P4: weight declared and justified, naive-versus-weighted contrast reported, and the one-line design statement written.
Milestone	M1 — Replication and interrogation	P4 enters M1 as Design-aware estimate . P4 is optional and ungraded with nothing to submit; M1 is where it is assessed.